Weather Intelligence & Disaster Alert System

Technical Document

Version 1.0.0

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1. Summary

Weather Intelligence & Alert System is a Smart Application designed to protect outdoor infrastructure. Its purpose is to support field operations and maintenance of high value equipment for the businesses. It is designed by ML Infomap to provide real-time and predictive weather insights and alerts to your desktop and mobile phone.

The Weather Intelligence & Alert System offers real-time monitoring, interactive visualization, weather hazard tracking, and automated communication— all from one Smart Application. The goal is simple: give users easy access to complex and dynamic weather information for their specific area and empower them with early alerts and actionable insights to protect lives and resources.

2. System Overview

2.1 System Snapshot

Attribute	Description
System Name	Weather Intelligence & Disaster Alert System
System Type	Real-time Monitoring, Prediction & Alert Platform
Primary Language	AngularJS(frontend), Flask(backend), Postgres(database)
Data Sources	WeatherAPI, VisualCross
Map Library	Open Layers
Notification Service	WhatsappAPI, smptblib for emails

2.2 Key Objectives

- Provide real-time data at circle and district levels.
- Deliver 7-day forecast data for circles and districts.
- Display weather hazards on a map, categorized by hazard type.
- Share circle-wise reports daily with the Indus team and structure-wise reports as needed.
- Issue cyclone alerts whenever detected.
- Enable KPI control for different hazards, reflected in reports.

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3. Application Portal Architecture

3.1 District Wise Weather

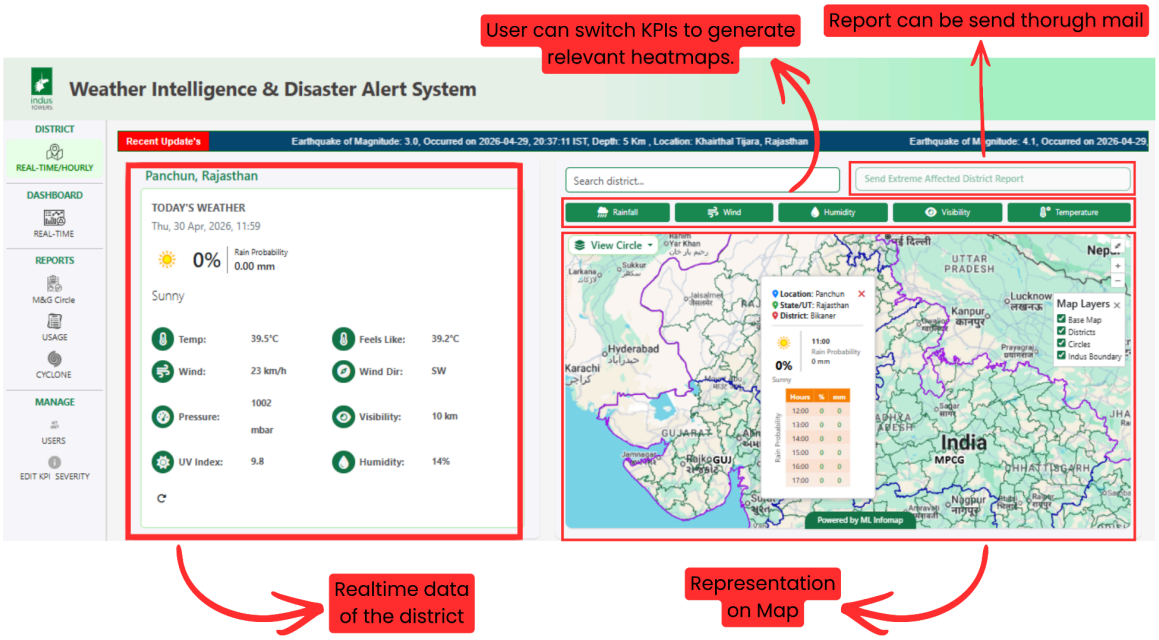


Fig. 3.1

- Displays current day/real-time weather data for a specified district, as the system provides real-time data at the district level.
- Features interactive visualization that allows the user to toggle between different Key Performance Indicators (KPIs).
- Creates a heatmap overlay on the map to reflect the data for the selected KPI, which enables KPI control for different hazards.
- Shows weather hazards on the map, which are categorized by the type of hazard.



Fig. 3.2

3.2 Circle Wise Weather

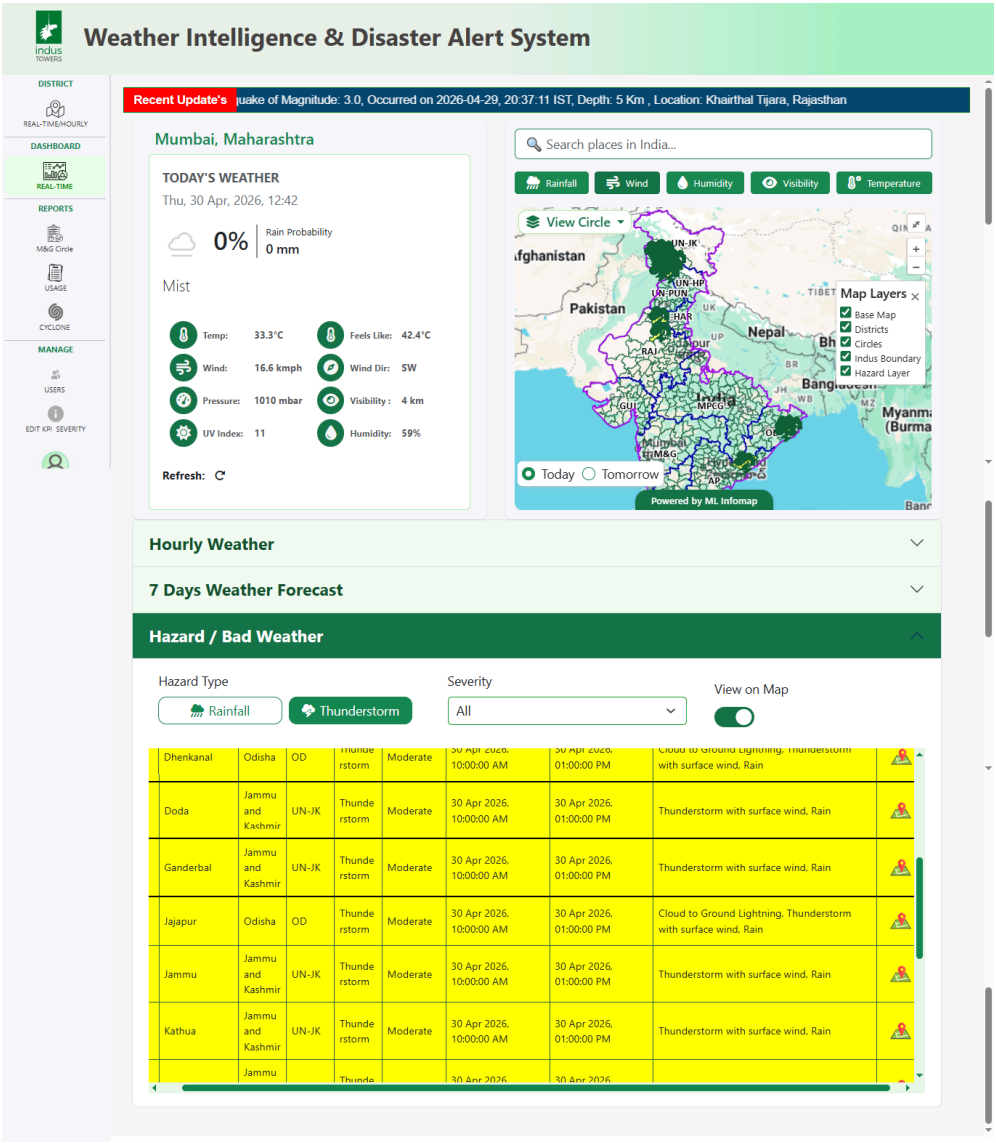


Fig. 3.3

- Circle-Level Data: Provides real-time weather data and daily reports specifically for the circle level.
- Hourly Updates: Utilizes sources like the NOAA National Weather Service, which are polled hourly for data ingestion.
- 7-Day Forecast: Delivers a 7-day weather forecast for circles.
- Bad Weather Alerts: Includes functionality to issue cyclone alerts whenever detected and display general weather hazards on the map.

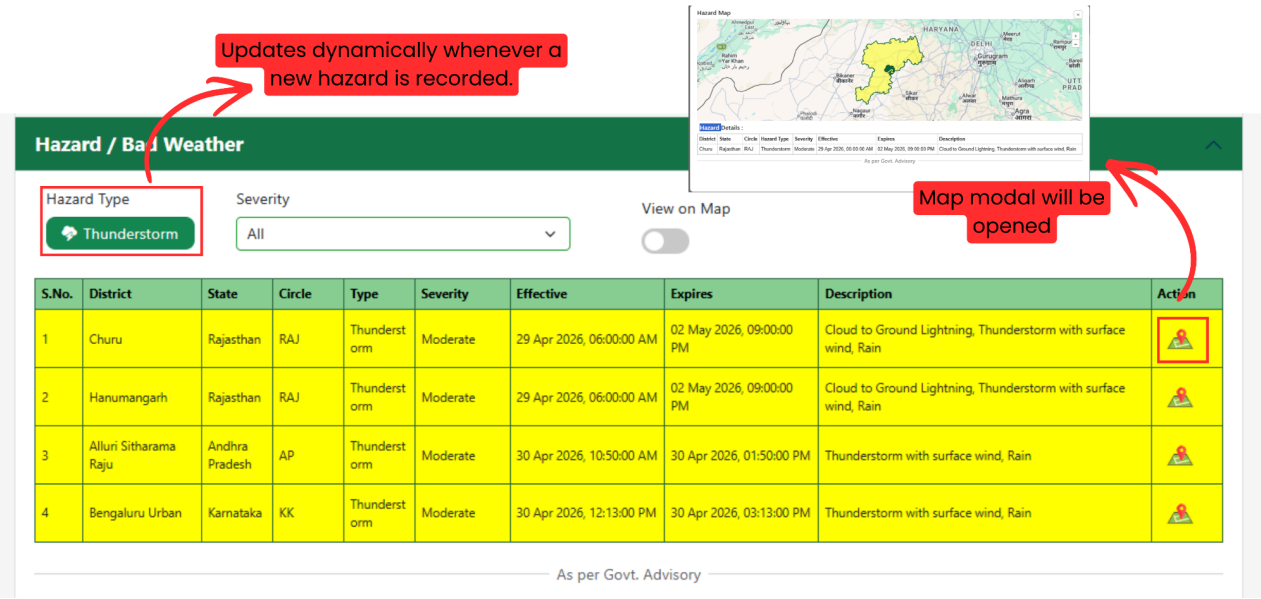


Fig. 3.4

3.3 Circle Wise Report

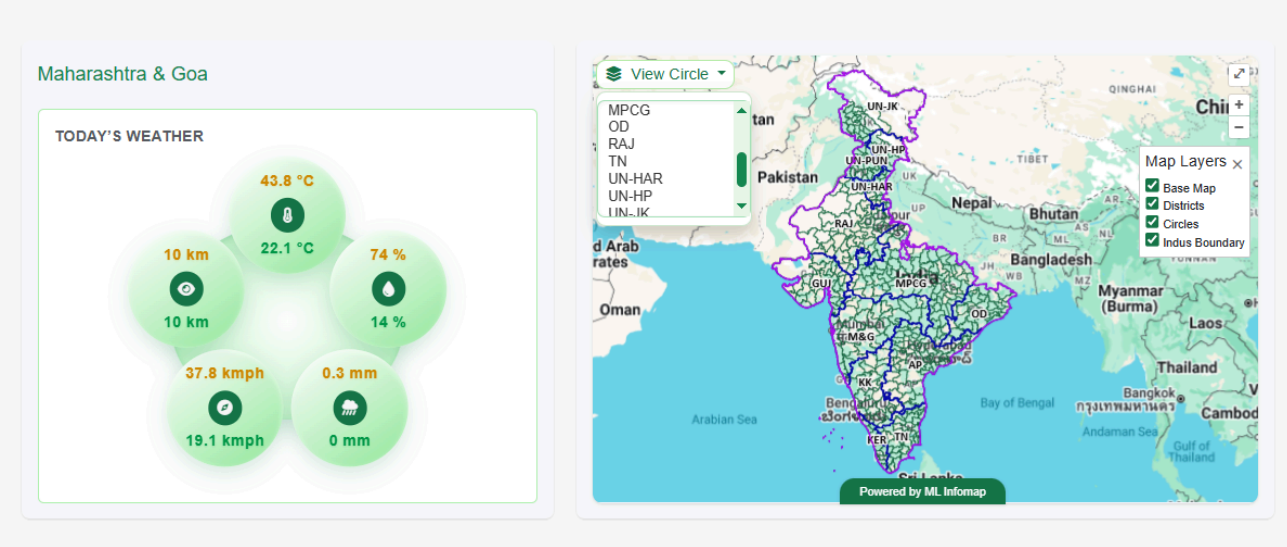


Fig. 3.5

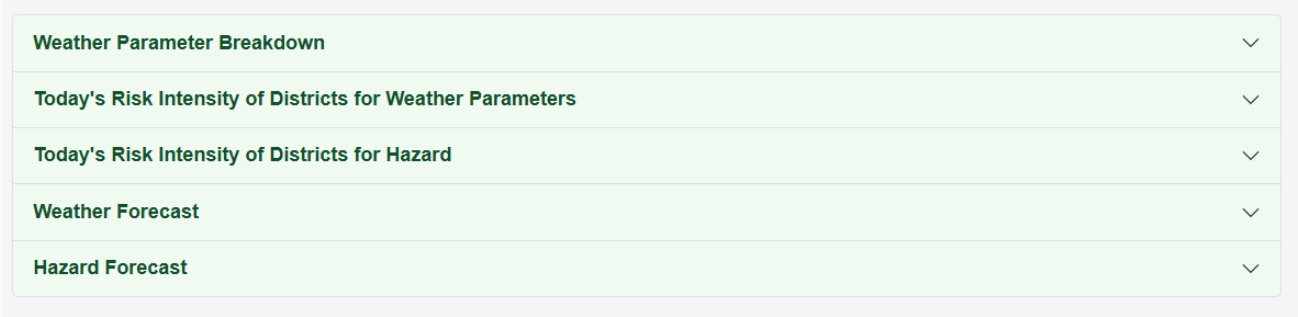


Fig. 3.6

- **Weather Parameter Breakdown:** Displays an overall breakdown of weather KPIs using a bar chart.
- **Today's Risk Intensity of Districts for Weather Parameters:** Displays district-wise risk levels for each weather parameter (Extreme, High, Moderate, Low).
- **Today's Risk Intensity of Districts for Hazard:** Displays district-wise hazard risk levels (Extreme, High, Moderate, Low).
- **Weather Forecast:** Provides district-wise weather data.
- **Hazard Forecast:** Provides district-wise hazard predictions.

3.4 Usage Report

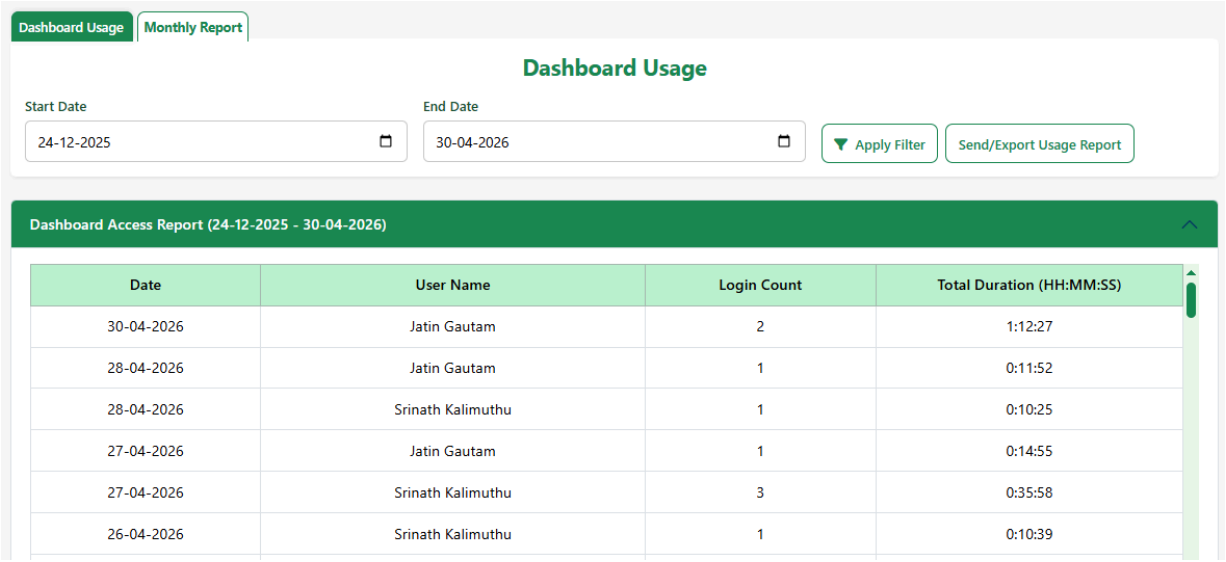


Fig. 3.7

- Captures user login events and monitors session duration.
- Sessions **automatically expire after 15 minutes of inactivity**, enforcing re-authentication.
- Restricts concurrent access by allowing **only one active session** per user at a time.

3.5 Cyclone Report

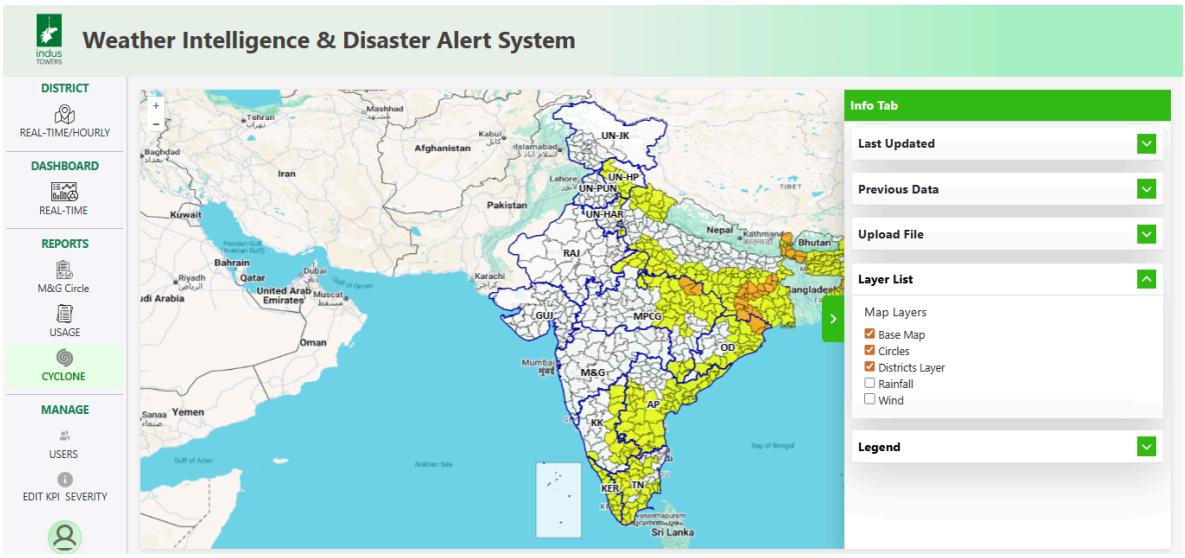


Fig. 3.8

- Data is sourced from the IMD website.
- The data is uploaded and processed within the portal.
- It is visualized on the map and made available as reports.
- The reports can also be shared via email when required.

3.5 User Management

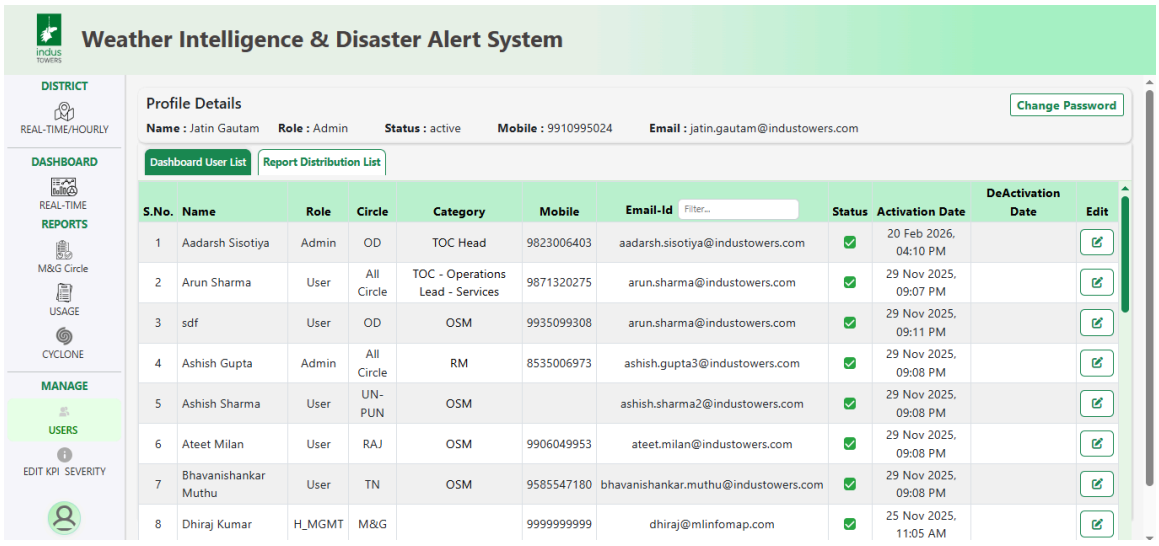


Fig. 3.9

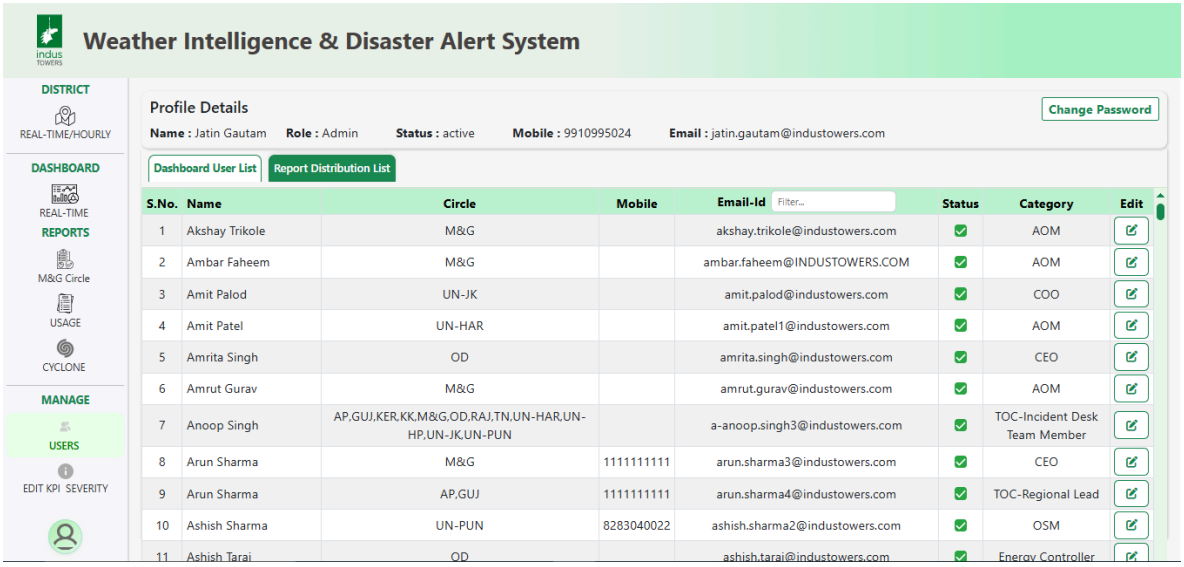


Fig. 3.10

- **In the Users tab, two buttons are available:**
  - **Dashboard User List :** Displays the list of users who have access to the dashboard. When this button is active, all user-related operations are performed on the Dashboard User List.
  - **Report Distribution List :** Displays the list of users who receive reports via Email . When this button is active, all user-related operations are performed on the Report Distribution List.

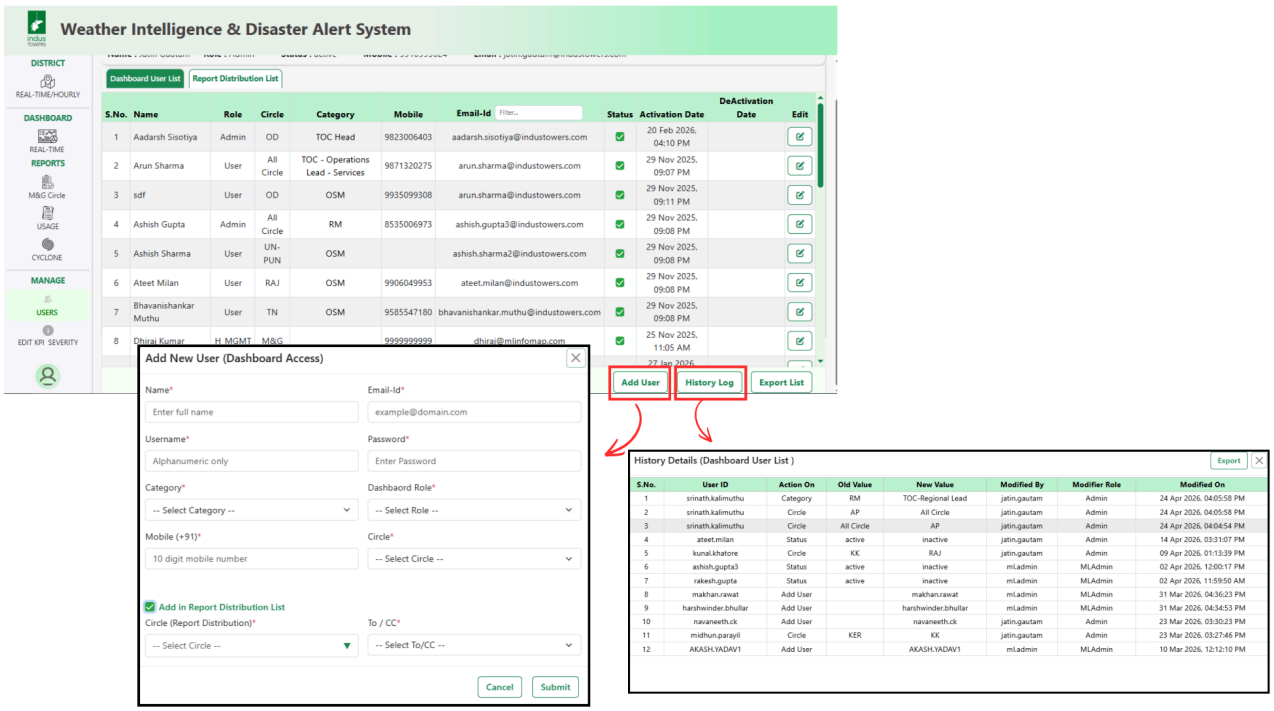


Fig. 3.11

- Admin can add users through the User Management section.
- All user-related actions are recorded in the history log.
- User status can be enabled or disabled as required.

3.5 Edit KPI Severity

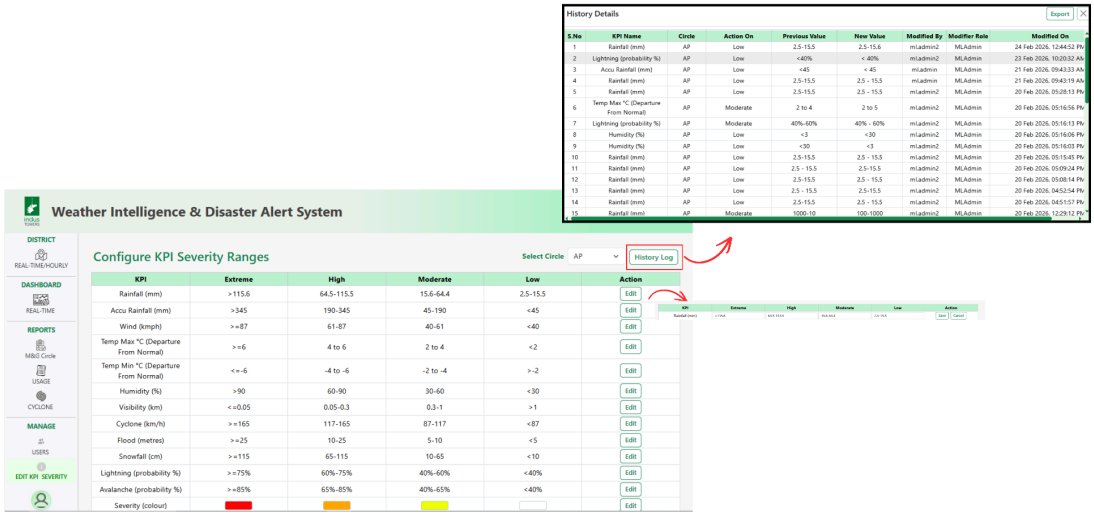


Fig. 3.12

- Admin can modify KPI severity in the KPI Severity section.
- Changes are reflected in the reports.
- A history log is maintained for all updates.

3.5 Hazard Feed

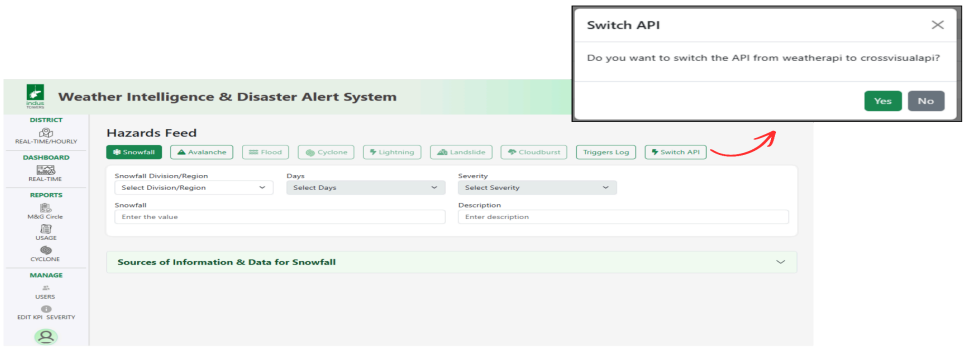


Fig. 3.12

WeatherAPI serves as the primary source for weather forecast data.

- In case of service unavailability, the system can switch to Cross Visual API as a backup source.

6. API Reference

All API endpoints are versioned under /api/. Authentication uses either a Bearer JWT (header: Authorization: Bearer <token>).

S.No	Module	Endpoint / API	Database Table	Purpose
1	Dashboard	https://api.weatherapi.com/v1/forecast.json	External — WeatherAPI.com	Current forecast, 24-hour hourly & 7-day forecast
2	Dashboard	get-ndma-hazards-list	disaster_ndma	Hazard types list for dashboard display
3	Dashboard	get-ndma-severity-list	disaster_ndma	Severity classifications for hazard filtering
4	Dashboard	get-ndma-today-disasters	disaster_ndma	Current-day disaster data for real-time monitoring
5	Dashboard	get-earthquake	earthquake_alerts	Earthquake alert data for marquee notification
6	District Monitoring	https://api.weatherapi.com/v1/forecast.json	External — WeatherAPI.com	Current, 24-hour hourly & 7-day weather forecast
7	District Monitoring	get_district_list	(circle-filtered query)	Districts for a selected circle
8	District Monitoring	send_district_weather_report	N/A — email dispatch	Send alert email with user & location payload
9	Circle Report	fetch_circle_report	district_wise_7dayfc_severity	Circle-level report data
10	Circle Report	fetch_kpi_legend_with_color	weather_kpi_controls	Severity colour config per KPI for charts & tables
11	Circle Report	fetch-kpi-range	weather_kpi_controls	KPI severity threshold ranges for legend & tables
12	Circle Report	fetch_district_wise_KPI_values	district_wise_7dayfc_severity	District KPI values by selected circle
13	Circle Report	fetch_accumulated_rainfall	district_wise_accum_rainfall	Accumulated rainfall data for selected circle
14	Circle Report	fetch_district_names_severity_wise	district_wise_7dayfc_severity	Districts grouped by severity level
15	Circle Report	get-hazard-affected-district	hazard_fog, hazard_snowfall, hazard_lightning, hazard_flood, hazard_cyclone, hazard_cloudburst, hazard_avalanche	Hazard data across all hazard category tables
16	Circle Report	get_circle_list	indus_circle_geomerty	Circle list bound to dropdown selector

S.No	Module	Endpoint / API	Database Table	Purpose
17	Circle Report	generate_pdf_link	N/A — PDF generation service	Generate downloadable PDF report URL
18	Circle Report	get_circle_weather_min_max	(KPI aggregation query)	Min/max KPI values with labels & icons
19	Circle Report	get-district-wise-hazards	hazard tables + district_geometry	District hazard data via spatial join
20	Usage	get_log_min_max_date	weather_user_activity_log	Min/max date range of usage records
21	Usage	get_log_summary_date_wise	weather_user_activity_log	Summarised usage data within date range
22	Usage	fetch_dashboad_usage	weather_user_activity_log	User-specific activity filtered by date & user ID
23	Usage	get_log_user_list	weather_user_activity_log	Users with activity records in date range
24	Usage	export_send_dashboard_usage	weather_user_activity_log	Export usage report or send via email
25	Cyclone	get_cyclone_geojson	cyclone_data_from_uploaded_file	Cyclone point & line layer data by date
26	Cyclone	get_indus_circle_boundary	indus_circle_geomerty	Circle boundary rendered as map layer
27	Cyclone	hazard_cyclone_report_data	hazard_cyclone	Cyclone report table data
28	Cyclone	get_india_level_districts	act_warning	District map layer with severity-based colour
29	Cyclone	get-current-weather	weather_hourly_data_all_india	Rainfall & wind IDW image layer data
30	Cyclone	fetch-kpi-range	weather_kpi_controls	Severity colours for district legend styling
31	Cyclone	send_cyclone_alert	N/A — email dispatch	Send cyclone alert email with PDF attachment
32	Cyclone	get_circle_list	indus_circle_geomerty	Circle list for impacted circle popup
33	Cyclone	cyclone_impacted_circles	cyclone_impacted_circles	Save selected impacted circles to DB
34	User Management	get_circle_list	indus_circle_geomerty	Circle options for user assignment
35	User Management	get-user-license	System license config	Total licensed user count allowed
36	User Management	get-user-list	licensed_user_auth	Dashboard user list with edit & add support
37	User Management	update_user_status	licensed_user_auth	Activate/deactivate user (with license check)
38	User Management	update_user	licensed_user_auth	Update existing user details
39	User Management	add_new_user	licensed_user_auth	Create new dashboard user record

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S.No	Module	Endpoint / API	Database Table	Purpose
40	User Management	update-user-license	weather_user_license	Update maximum allowed user count
41	User Management	change-user-password	licensed_user_auth	Update user password by email
42	User Management	get_report_user_list	master_users	Report distribution user list
43	User Management	get_users_category	master_users	Category dropdown for add-user form
44	User Management	update_report_user_status	master_users	Activate/deactivate report distribution user
45	User Management	update_report_user	master_users	Edit report distribution user details
46	User Management	add_new_report_user	master_users	Add new report distribution user
47	User Management	get-user-history	user_management_history	User activity history log
48	User Management	get-user-report-circles	master_users	Circle assignments for a specific user
49	KPI Severity	fetch-kpi-range	weather_kpi_controls	KPI severity range values (editable table)
50	KPI Severity	get_circle_list	indus_circle_geomerty	Circle dropdown for KPI selection
51	KPI Severity	update-kpi-range	weather_kpi_controls	Save updated KPI ranges for selected circle
52	KPI Severity	get-kpi-history	kpi_modify_history	KPI modification history with change details

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